

FRUIT AND COLOR USING BACKTRACKING

AIM

To write a Prolog program to identify fruits and their colors using backtracking.

ALGORITHM

1. Start.
2. Define fruit and color facts.
3. Associate each fruit with its color.
4. Query the database for a fruit or color.
5. Prolog uses backtracking to find all matching solutions.
6. Display the results.
7. Stop.

SOURCE CODE

% Fruit and Color Database

fruit_color(apple, red).

fruit_color(banana, yellow).

fruit_color(orange, orange).

fruit_color(grapes, green).

fruit_color(mango, yellow).

fruit_color(guava, green).

OUTPUT

```
?- fruit_color(Fruit, yellow).  
Fruit = banana ;  
Fruit = mango.  
  
?- fruit_color(mango, Color).  
Color = yellow.  
?-
```

RESULT

The Prolog program successfully identifies fruits and their colors using backtracking to retrieve multiple matching solutions.